For the exclusive use of C. Su, 2026.

IN2088

INSEAD The Business School for the World®

Case Study

DeepSeek:

A Chinese Dark Horse in the Global AI Arena

08/2025-6993

This case study was written by Yi Tang, Associate Professor of Strategy at University of Hong Kong, Yuanhao Li, Research Assistant at the University of Hong Kong, and Guoli Chen, Professor of Strategy and Mubadala Chaired in Corporate Governance and Strategy at INSEAD. It is intended to be used as a basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

To access INSEAD teaching materials, go to https://publishing.insead.edu/.

Copyright © 2025 INSEAD. All rights reserved.

NO PART OF THIS PUBLICATION MAY BE COPIED, STORED, TRANSMITTED, TRANSLATED, REPRODUCED OR DISTRIBUTED IN ANY FORM, MEDIUM, APPLICATION, OR SYSTEM WITHOUT THE PERMISSION OF THE COPYRIGHT OWNER. SUCH RESTRICTION SHALL ALSO COVER AI SYSTEMS AND AI MODELS.

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

On January 27, 2025, the US stock market opened with a sharp decline, the technology sector being hit particularly hard. The plunge reflected the recent breakthrough of the Chinese AI startup DeepSeek, which challenged the "invincibility" of the US tech sector. The US chip giant Nvidia saw its stock price plummet 17%, while Broadcom dropped 17%, AMD fell 6%, and Microsoft declined 2%. Companies in AI-related industries such as power suppliers also suffered heavy losses. American Electric Power saw its stock price tumble 21%, while Vistra experienced a 29% drop.[1] Meanwhile, DeepSeek shot to the top of the App Store's free app rankings in the United States, surpassing OpenAI's ChatGPT, and in China was ranked the No. 1 free app on the App Store.[2]

In an interview on January 24th, the founder of data annotation company Scale AI, Alexandr Wang, stated that DeepSeek performed the best in its tests, matching the top generative AI models in the United States. He had previously warned that DeepSeek-V3 was a "lesson" to the US: "While Americans rested, China (tech) worked, catching up at lower cost, faster speed, and with greater strength."[3] After the launch of its open-source model DeepSeek-V3 on December 27, 2024, it was ranked seventh by LMArena among all models, first among open-source models, and the most cost-effective of the top ten models in the world. The cost of training DeepSeek-V3 was $5.576 million, a fraction (1% to 5%) of the training costs of similar LLMs from companies like OpenAI.[4] Less than a month later, on January 20, 2025, DeepSeek officially open-sourced its R1 inference model, which meant anyone could distill R1 to train other models under the MIT License.[5]

By January 24, DeepSeek-R1 ranked third on Chatbot Arena's leaderboard, tying with the top inference model o1. In areas demanding extremely high technical proficiency—such as advanced prompting, coding, and mathematics – DeepSeek-R1 took first place. In terms of style control, DeepSeek-R1 tied with o1 in first place, indicating that the model excelled at understanding and following user instructions while generating content in a specified style. In tests combining advanced prompts with style control, DeepSeek-R1 once again tied with o1 for first place, demonstrating its ability to handle complex tasks with great precision.[6]

Liang Wenfeng: From High-Flyer Quant to DeepSeek

Hangzhou Deep Exploration AI Fundamental Technology Research Co., Ltd was founded by Liang Wenfeng in July 2023, dedicated to in-depth research and development of large language models (LLM) and related technologies. According to public records, Liang was born in 1985 in Zhanjiang, Guangdong, China.

1 Houfu Capital, "What Is the 'DeepSeek' Everyone Is Talking About?," NetEase, 2025, https://www.163.com/dy/article/JN5DU7HI0517CIB2.html, accessed March 2025.
2 CNBC, "China's DeepSeek AI Tops ChatGPT in App Store: What You Should Know," CNBC, 2025, https://www.cnbc.com/2025/01/27/chinas-deepseek-ai-tops-chatgpt-app-store-what-you-should-know.html accessed March 2025.
3 Alexandr Wang, personal profile, X, 2025, https://x.com/alexandr_wang, accessed March 2025.
4 "Over the Weekend, DeepSeek Dropped a 'Bombshell' by Disclosing Cost-Profit Margins for the First Time – How Significant Is the Impact?," Sina Finance, March 2, 2025, https://finance.sina.com.cn/stock/zqgd/2025-03-02/doc-inenfmip3745430.shtml, accessed March 2025.
5 The MIT License is a permissive software license originating at the Massachusetts Institute of Technology (MIT) in the late 1980s. As a permissive license, it puts very few restrictions on reuse and therefore has high license compatibility.
6 Daily Economic News, "Silicon Valley Shocked! Chinese AI Dominates Foreign Media, AI Leaders Say: 'It Has Caught Up with the U.S.!' Silicon Valley Engineers Work Overnight to Replicate, DeepSeek May Completely Change the Game," Baidu Baijiahao, 2025, https://baijiahao.baidu.com/s?id=1822291558785861433, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

At the age of 17 he was admitted to Zhejiang University, where he majored in electronic information engineering and later pursued a master's degree in information and communication engineering, focusing on machine vision research under Professor Xiang Zhiyu.[7] In 2008, after graduating with a degree in Software Engineering, his chosen path differed from his peers, who typically joined large corporations as programmers. Driven by a vision of integrating wealth/finance with technology, he chose quantitative investment, an emerging field in China characterised by uncertainty and skepticism at that time. He started High-Flyer Quant. Many doubted whether it could be profitable – relying on computer algorithms and data models seemed abstract and unreliable. Liang, however, was captivated by the underlying scientific logic. In 2010, the launch of the CSI 300 Index Futures opened up unprecedented opportunities for the development of quantitative investment.[8] [9] With a solid technical foundation and keen market insights, his High-Flyer Quant made remarkable strides. Their quantitative investment strategies achieved tremendous success and proprietary funding snowballed to exceed 500 million yuan. Liang made a name for himself in the field of quantitative investment, laying the foundation for his future endeavors.

As deep learning algorithms achieved significant breakthroughs globally, artificial intelligence (AI) saw explosive development. In 2015, Liang started working with several like-minded Zhejiang University alumni whose ambition was to use the powerful tools of mathematics and AI to create a world-class quantitative hedge fund. Jiuzhang Asset was established in Hangzhou on June 11, 2015, and High-Flyer Quant was founded in Ningbo on February 15, 2016, with Liang as the controller of both.

Within a year of founding High-Flyer Quant, they had launched the first AI-driven live trading system, leveraging their sophisticated understanding of technology and extensive market experience, while sending ripples through the financial markets. As their research and application of AI technologies increased, they gradually transformed all their trading strategies into AI-driven systems. The returns on High-Flyer Quant's funds grew significantly, outperforming the CSI 300 Index over the same period and attracting investor attention. A tidal wave of capital flowed into High-Flyer Quant, propelling the fund's size to new heights. By 2021, it had had successfully surpassed the 100 billion yuan mark, ranking among the top four quantitative investment firms in China and one of the leading enterprises in the domestic field.[10] [11]

However, as High-Flyer Quant's fund size continued to expand, it came up against a major challenge: exponential growth in demand for computational power to underpin its AI trading strategies. To meet the need and ensure the company remained competitive, Liang made a bold decision—to make large-scale

---

7 "Liang Wenfeng," *Wikipedia*, last modified March 5, 2025, https://zh.wikipedia.org/wiki/梁文锋, accessed March 2025.

8 "DeepSeek's Billionaire Hedge Fund Founder – Liang Wenfeng: A Legendary Journey from Quantitative Investment to AI Pioneer (Including Growth Experience & Public Statements)," Sina Finance, February 9, 2025, https://finance.sina.com.cn/money/fund/etf/2025-02-09/doc-ineivtuk8308749.shtml, accessed March 2025.

9 "Liang Wenfeng (Founder of DeepSeek)," Baidu Baike, https://baike.baidu.com/item/%E6%A2%81%E6%96%87%E9%94%8B/65336579, accessed March 2025.

10 "The Billionaire Behind DeepSeek – Liang Wenfeng: A Legendary Journey from Quantitative Investment to AI Pioneer (including growth experience & public statements)," Sina Finance, February 9, 2025, https://finance.sina.com.cn/money/fund/etf/2025-02-09/doc-ineivtuk8308749.shtml, accessed March 2025.

11 "Founder of Quantitative Giant Phantom Square, Liang Wenfeng, Attends Premier's Symposium and Speaks—He Also Founded the 'Pinduoduo of AI'," The Paper (Pengpai News), January 2025, https://baijiahao.baidu.com/s?id=1821904728764706575&wfr=spider&for=pc, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

investments in AI computational power. In 2019, High-Flyer Quant invested RMB 200 million ($ 27.4 million) to build the "Firefly One" AI computing cluster, equipped with 1,100 GPU[12] cards.

When Nvidia released its A100 chips, Liang once again demonstrated foresight, being one of the first in Asia-Pacific region to secure the chips. In 2021, High-Flyer Quant invested another 1 billion yuan to build "Firefly Two", which was equipped with 10,000 A100 GPU cards, boasting an astonishing level of computational power equivalent to 760,000 personal computers. Its physical footprint exceeded the size of 10 basketball courts.

In the era before the boom of large AI models, such a massive investment sparked criticism and skepticism. Some argued that it was unrealistic for a private equity firm to spend such a vast amount on accumulating computational resources that were unlikely to generate immediate economic benefits in the short term. But Liang remained steadfast, understanding that computational power was the cornerstone of future technological competition, particularly in the field of AI, where it was the foundation of algorithm training and optimization. Indeed his large-scale investment in computational power laid the groundwork for High-Flyer Quant's subsequent exploration and innovation in AI, providing indispensable support for the development of DeepSeek-V3 and DeepSeek-R1.[13]

DeepSeek

Founded in 2023, DeepSeek marked Liang's entry into the field of artificial general intelligence (AGI). It employed 139 engineers and researchers (including the founder). In comparison, OpenAI (the developer of ChatGPT) had 1,200 researchers. Anthropic (developer of the Claude model) had more than 500.

As of 2025, Liang was the sole major shareholder and primary financial backer of DeepSeek. Within Hangzhou Deep Exploration he directly and indirectly held 84.29% of the shares and nearly 100% of the voting rights. DeepSeek had no recorded external financing as of 2025. Personally and through his controlled entities, he was the primary investor, bearing almost all the financial burden. He told an interviewer: "(DeepSeek) has no short-term plans for financing. Our problem has never been money but rather the ban on high-end chip exports."[14]

People and Passion

A factor in DeepSeek's success was its technological "idealism"—it pursued tech advancement for its own sake, rather than chasing new applications and monetization, as Liang explained:

> *"…because we believed the most important thing at that moment was to participate in the global wave of innovation. For many years in the past, Chinese companies were accustomed to taking technologies innovated by others and monetizing them through applications. But*

12 A GPU (Graphics Processing Unit) is a specialized electronic circuit designed to rapidly manipulate and alter memory to accelerate the creation of images in a frame buffer intended for output to a display device.

13 "The Billionaire Behind DeepSeek – Liang Wenfeng: A Legendary Journey from Quantitative Investment to AI Pioneer (including growth experience & public statements)," Sina Finance, February 9, 2025, https://finance.sina.com.cn/money/fund/etf/2025-02-09/doc-ineivtuk8308749.shtml, accessed March 2025.

14 "I Advised Liang Wenfeng Many Times: DeepSeek Needs Financing," Sina Finance, February 10, 2025, https://finance.sina.com.cn/roll/2025-02-10/doc-ineiytxe3609279.shtml, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

> *this was not something that should have been taken for granted. In this wave, our starting point wasn't merely to seize the chance to make quick profits but rather to move toward the technological frontier and actively contribute to the entire ecosystem. We believed that with economic development, China should gradually become a contributor rather than always being a free-rider. During the past three decades of the IT revolution, we had hardly participated in true technological innovation. We had grown accustomed to Moore's Law falling from the sky – comfortably waiting at home for better hardware and software to appear every 18 months."*[15] [16]

Traditional enterprises relied on "qualifications" to filter talent, but DeepSeek adopted an unconventional recruitment method: it ignored qualifications and experience, instead selecting young people who genuinely loved the profession. While most domestic companies lamented the shortage of top AI talent, Liang argued that finding experienced candidates was only a matter of urgency if one had short-term goals – in this respect China was behind Silicon Valley at the time – but from a long-term perspective, experience ceased to be critical. Instead, "fundamental ability, creativity, and passion" were the core qualities, hence there was no shortage of suitable candidates within China.

Its core team primarily consisted of fresh graduates and researchers with two or three years of work experience. Rather than rush to recruit star experts from abroad, it hired doctoral students and researchers who had recently graduated from China's top universities, giving them ample resources and independence to initiate their own projects. As long as they could demonstrate the feasibility of their idea, they were given access to massive amounts of GPU computing power for large-scale training and were free to form cross-team collaborations. In this way, some newly-hired young employees grew into technical project leaders within a few months. They generated fresh ideas through practical experience.[17]

DeepSeek believed that pure passion and intrinsic curiosity could overcome the limitations imposed by experience, leading to a flatter management structure. The working environment was described as "extremely free," characterised by minimal bureaucratic culture or the pressure of KPIs. People were encouraged to "conduct research they truly wanted to pursue," believing in the notion that, "Innovation was rarely arranged, and certainly not taught through KPI assessments. Rather, it originated from bottom-up passion and creative thinking."[18] DeepSeek abandoned the traditional KPI evaluation system. Unlike other enterprises that set fixed goals, DeepSeek allowed its researchers to freely explore the most challenging problems, making passion—not metrics—the main driver of innovation.

Within this unique environment, even the CEO spent most of his working day reading research papers and coding. There was no rigid hierarchy, only a natural division of labor based upon shared ideals. The MLA[19]

---

15 "Shocking the AI World: How DeepSeek Used 150 People to Launch a Disruptive Organizational Revolution" Sina Finance, February 6, 2025, https://finance.sina.com.cn/cj/2025-02-06/doc-ineiphxm1176589.shtml, accessed March 2025.

16 "Interview with Liang Wenfeng of High-flyer: The 'Pinduoduo of AI'—Unveiling DeepSeek!" WallStreetCN, July 18, 2024, https://wallstreetcn.com/articles/3719982, accessed March 2025.

17 "Shocking the AI World: How DeepSeek Used 150 People to Launch a Disruptive Organizational Revolution" Sina Finance, February 6, 2025, https://finance.sina.com.cn/cj/2025-02-06/doc-ineiphxm1176589.shtml, accessed March 2025.

18 "Liang Wenfeng: DeepSeek Has No KPIs." NetEase, https://www.163.com/dy/article/JP8I0CKK05118I96.html.， accessed April 2025.

19 The main purpose of proposing MLA (Multi-Level Attention) is to reduce the KV Cache during inference, thereby enabling longer context processing on fewer devices, or increasing batch size with the same context length, resulting in faster inference speeds or greater total throughput, ultimately reducing inference costs.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

architecture at the core of DeepSeek V3 served as a prime example. It began as a researcher's side-interest, and with the team's support eventually evolved into a critical breakthrough that significantly improved AI model efficiency and reduced costs.[20]

DeepSeek had almost no explicit administrative hierarchy. Managers primarily served to allocate resources from the top whenever an idea demonstrated potential. As for Liang, despite being the boss, he was deeply involved in frontline work every day—writing and running code. His colleagues remarked that he "didn't seem like a boss at all, but rather more like a true geek."[21] "Innovation required minimal intervention and management, allowing everyone the freedom and opportunity to experiment and make mistakes. Innovation often emerged spontaneously, rather than being deliberately arranged or explicitly taught."[22] With only 150 employees, the company replaced the traditional hierarchical structure with an extremely flat management model where each talented individual could unleash their creativity to the fullest. DeepSeek organized work in a bottom-up manner, without assigning predefined roles or positions. A division of labor naturally emerged as people proactively undertook tasks based on their own experience and ideas. When team members encountered challenges, they invited others to join the discussion. Once an idea proved to have development potential, the company would allocate substantial resources from the top to accelerate its progress.

Products (LLMs) of DeepSeek [see Exhibit 1]

Although DeepSeek started later than other large language model companies, it benefited from strong computing resources and deep expertise. Throughout 2024 it experienced rapid growth, launching DeepSeek LLM, DeepSeek-V2 and DeepSeekV3 in quick succession, and in January 2025 it launched DeepSeek-R1.

The corresponding papers published were "DeepSeek LLM: Scaling Open-Source Language Models with Long-termism", "DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model", "DeepSeekV3 Technical Report", and "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning".[23] For the DeepSeek LLM model released in January 2024, the corresponding paper primarily covering model training, scaling law analysis, alignment optimization, and evaluation work. From this project, researchers successfully leveraged small-scale experiments to predict the performance of large-scale models accurately, which provided valuable guidance for future model scaling directions, enabling more rational resource planning during model development and improved performance.

In May 2024, DeepSeek released DeepSeek-V2, which quickly gained popularity due to its innovative model architecture and unprecedented cost-effectiveness. The API pricing[24] of DeepSeek-V2 was set at 1

20 "Shocking the AI World: How DeepSeek Used 150 People to Launch a Disruptive Organizational Revolution" Sina Finance, February 6, 2025, https://finance.sina.com.cn/cj/2025-02-06/doc-ineiphxm1176589.shtml, accessed March 2025.

21 "The Legend Behind: DeepSeek Founder Liang Wenfeng's Innovation Insights," Sina Finance, February 3, 2025, https://finance.sina.com.cn/tech/2025-02-03/doc-ineiefek7393482.shtml, accessed March 2025.

22 "Liang Wenfeng: DeepSeek Has No KPIs." NetEase, https://www.163.com/dy/article/JP8I0CKK05118I96.html.， accessed April 2025.

23 "Overview of DeepSeek's Background and Preliminary Exploration of Its Applications in the Financial Sector," Sohu Finance, February 28, 2025, https://www.sohu.com/a/865272147_121123901, accessed March 2025.

24 API pricing in large language models refers to the cost structure implemented by AI providers for accessing their models through application programming interfaces. It typically involves charges based on input and output tokens processed, with different models having varying rates depending on their capabilities and computational requirements.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

RMB ($ 0.14) per million input tokens, and 2 RMB ($0.28) per million output tokens—at just 1% of the price of GPT-4 Turbo. It was a powerful mixture-of-experts (MoE) language model, featuring an innovative architecture based on the Transformer design and incorporating Multi-head Latent Attention (MLA).

On December 27, 2024, DeepSeek released DeepSeek-V3, an MoE language model with 671 billion parameters. The research focused on model architecture, training methodologies, evaluation and application scenarios dedicated to enhancing open-source model performance and pushing language models toward artificial general intelligence (AGI). Within this work, Multi-Token Prediction aimed to enhance model performance by prompting the model to predict multiple future tokens simultaneously at each position. The FP8 mixed-precision training framework aimed to improve computational efficiency, reduce memory usage, and lower training costs by decreasing data precision, all while maintaining model accuracy and performance. In evaluations it not only surpassed top-tier open-source models such as Qwen2.5-72B (Alibaba's LLM) and Llama 3.1-405B (Meta's LLM), but its performance was comparable to leading closed-source models like GPT-4o and Claude 3.5-Sonnet (Anthropic's LLM).

Alongside the release and simultaneous open-sourcing of the DeepSeek-V3 model, a 53-page document detailed the training procedures and technical specifics. The substantially upgraded V3 model was trained with an "unimaginably" tight budget: the entire cost was 5.576 million USD, and was completed over 55 days on a cluster of 2,048 Nvidia H800 GPUs (a lower-performance variant tailored for the Chinese market)—less than one-tenth of the cost of training OpenAI's GPT-4o model.

The subsequent DeepSeek-R1demonstrated that the reasoning capabilities of LLMs could be significantly enhanced through reinforcement learning. It explored using model distillation techniques to enhance the reasoning ability of smaller models. Its performance was comparable to ChatGPT-o1, but at a significantly lower training cost, primarily due to its innovative training algorithm design, hardware-optimized attention mechanisms, and efficient data utilization strategies.

Open-source vs. Closed Source AI Models

Open-source models are AI models whose source code, model parameters, training methods and related data are made publicly available, allowing anyone to freely access, use, modify, and redistribute them.[25] They offer high transparency and flexibility, enabling users to fine-tune and customize them according to their needs. Examples included Meta’s Llama series, Alibaba’s Qwen series, and the DeepSeek series. Among their many advantages, they foster innovation, reduce R&D costs, and enhance data security (through local deployment). However, users must invest their own hardware and tech expertise for their deployment and maintenance.

Closed-source models do not disclose their code or model parameters and provide only API access. Users pay to access functionalities through subscription or pay-as-you-go plans, but are not able to view or modify the internal implementation details and parameters. Examples include OpenAI’s GPT series, Anthropic’s Claude series, and Google’s Gemini series. Closed-source models deliver strong and stable performance and are simpler and more convenient for users to operate, but lack transparency and

25 "Open-Source Artificial Intelligence," Wikipedia, last modified March 8, 2025. https://en.wikipedia.org/wiki/Open-source_artificial_intelligence, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

customization capabilities. They typically require users to upload their data to third-party servers, presenting a potential risk to data privacy and security.[26]

When asked: "DeepSeek is a commercial organization, not a non-profit research institution – if you choose to innovate and then share it through open-source, where will the competitiveness be? For example, with the MLA architecture innovation in May, won't others copy it quickly?" Liang replied that in the face of disruptive technologies, the moat created by closed-source was only temporary – even if OpenAI kept their models closed-source, it could not prevent others from catching up:

> *"So, we place our value in the team. Our colleagues grow through this process, accumulate a lot of know-how, and form an organization and culture capable of innovation—this is our moat. Open-sourcing and publishing papers don't cause any real loss. For technical personnel, being followed is a rewarding experience. Open-sourcing is more like a cultural act rather than a commercial one. Giving is, in a sense, an additional honor. A company that does this also gains cultural appeal."*[27]

On February 24, 2025, DeepSeek launched "Open Source Week", announcing that it would open-source five code libraries to share its research progress with the global developer community in a fully transparent manner. New content was unlocked daily, sharing its advancements.[28]

On day 1, the first open-sourced code library was FlashMLA. Described as an "accelerator" for maximizing GPU potential, this was DeepSeek's high-efficiency MLA decoding kernel for Hopper GPUs, optimized for variable-length sequences and already in production. FlashMLA was specifically designed to optimize the computational efficiency of GPUs, particularly those under Nvidia.[29] [30]

DeepSeek announced the open-sourcing of DeepEP on Day 2, the first open-source EP communication library for MoE model training and inference. Known as the "communication steward" for large model training, DeepEP was designed to improve training efficiency for large models. For example, when multiple AI expert models (MoE architecture) worked together, it efficiently coordinated communication between them, reducing latency and resource consumption. It also supported low-precision computation (such as FP8), further saving computational power.[31]

On Day 3, DeepSeek announced the open-sourcing of DeepGEMM, described as the "energy-saving expert" for matrix computation. The tool was optimized for matrix multiplication, a core computation in AI training.[32]

---

26 "Open-Source Artificial Intelligence," Wikipedia, last modified March 8, 2025. https://en.wikipedia.org/wiki/Open-source_artificial_intelligence, accessed March 2025.

27 "Interview with Liang Wenfeng of Phantom Square: The 'Pinduoduo of AI'—Unveiling DeepSeek," WallStreetCN, July 18, 2024, https://wallstreetcn.com/articles/3719982, accessed March 2025.

28 Li Xiaoyin, "Five Days, Five Explosions! A Recap of DeepSeek's 'Open Source Week' Tech Package: Redefining Computing Power with Software!" WallStreetCN, February 28, 2025, https://wallstreetcn.com/articles/3742093, accessed March 2025.

29 Ibid.

30 Wang Wei, "What Magic Box Did DeepSeek Open Source Week Unlock This Time? How Will It Impact AI Development?", Sina Finance, March 1, 2025, https://finance.sina.com.cn/jjxw/2025-03-01/doc-inencaau9247638.shtml, accessed March 2025.

31 Ibid.

32 Wang Wei, "What Magic Box Did DeepSeek Open Source Week Unlock This Time? How Will It Impact AI Development?", Sina Finance, March 1, 2025, https://finance.sina.com.cn/jjxw/2025-03-01/doc-inencaau9247638.shtml, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

DeepSeek open-sourced two tools and a dataset on Day 4: DualPipe, EPLB, and analysis data from its training and inference frameworks. DualPipe was designed to address the "waiting time" issue in pipeline parallelism. For example, when multiple task steps had varying speeds, it enabled bi-directional scheduling to reduce idle time. EPLB was designed to automatically balance GPU workloads. When certain AI expert model tasks were overloaded, it replicated tasks to idle GPUs, avoiding a situation where "some are overworked while others are underutilized."[33]

On Day 5, DeepSeek announced the open-sourcing of the 3FS (Fire-Flyer File System). Described as the "speedy combo" for data processing, 3FS utilized a distributed file system leveraging high-speed storage and network technologies (such as SSD[34] and RDMA[35]), achieving data read speeds of up to 6.6 TB per second, making it ideal for massive data training applications. [36]

"DeepSeek's five-day wave of open-sourcing was equivalent to building a floating bridge over Nvidia's AI moat," said Tan Jian, associate professor of intelligent interaction design at Beijing University of Posts and Telecommunications, in an interview with The Paper. More importantly, the open-source modules from the DeepSeek team demonstrated their ability to deeply analyze Nvidia's CUDA and the tightly coupled model of parallel computing chips which was considered an unshakable software and hardware infrastructure in the AI research field, and represented the broad moat of Nvidia. Tan believed that Open Source Week's intensive release of open-source models and algorithms reconstructed the operational logic of AI hardware. It not only addressed previous doubts about DeepSeek's model training requiring immense computing power, but also showed potential for invigorating innovation among global AI software and hardware teams.[37]

Some felt that this open-sourcing by DeepSeek could be regarded as a “moral compass of the industry". While it had limited direct application for ordinary users and most end-users, it was extremely valuable for those working on foundational technologies. DeepSeek open-sourced all infrastructure optimizations mentioned in the DeepSeek-V3 paper. Many open-source frameworks could integrate these optimizations, and as hardware resources were once again compressed, this might lead to a wave of API price reductions, continuing to steer the industry toward openness and transparency.

Industry & Competition Overview

The AI Industry

The AI industry encompassed the entire industrial chain related to AI technologies, including fundamental research, hardware devices (such as AI chips and servers), software development platforms (such as machine learning frameworks and development tools), and various industry-level application solutions. It involved numerous technical fields – machine learning (ML), deep learning (DL), natural language processing (NLP), computer vision (CV), speech recognition, intelligent robotics, and many other branches.

33 Ibid.
34 SSD (Solid State Drive) is a storage device using flash memory for faster data access than traditional hard drives.
35 RDMA (Remote Direct Memory Access) is a technology that allows computers to exchange data directly between memories without CPU involvement, reducing latency for high-performance computing and networking.
36 Ibid.
37 "Building a Floating Bridge to Breach NVIDIA’s Moat? DeepSeek Open Source Week Shakes the Industry, China’s AI Overtaking on the Curve," NetEase News, March 1, 2025, https://www.163.com/dy/article/JPJHGLVK05118HNL.html, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

These fields were relatively independent yet interconnected, collaboratively forming a comprehensive and diversified AI ecosystem.[38]

In 2023, the global AI market was estimated to be worth USD 196.63 billion and was projected to grow at a compound annual growth rate (CAGR) of 36.6% between 2024 and 2030, signaling a period of rapid expansion and accelerating scale. The trend was driven by continuous advances in machine learning algorithms, the widespread availability of big data, and substantial improvements in computing power. Collectively, these facilitated breakthroughs in AI technologies, enabling remarkable progress in innovation and the commercialization of new AI-enabled applications.[39] [see **Exhibit 2**]

Meanwhile, the global technology giants and leading enterprises intensified their investments in research and innovation, actively promoting the adoption and integration of AI technologies across various industry verticals such as automotive, healthcare, retail, finance, and manufacturing. These gradually implemented more advanced and efficient technological solutions, improving operational efficiency and giving rise to new business models. As the expansion of AI technologies accelerated and innovative applications continued to emerge, AI disrupted traditional operational models and simultaneously created entirely new business segments and market opportunities. This further accelerated the overall development and evolution of the AI industry.

Another characteristic of the AI market was the high frequency of mergers and acquisitions (M&A), particularly by market-leading companies that actively pursued strategic acquisitions to gain access to cutting-edge AI technologies, talent pools, and integrated industry resources. Through strategic mergers, companies sought to consolidate their market position and achieve economies of scale in an increasingly competitive and rapidly growing market. Among the multiple motivating factors were the intense competition for technological capabilities and specialized talent, the growing strategic importance of AI technology, and a heightened sense of urgency to establish advantageous market positions through strategic acquisitions.

Continuous technological innovation, the expansion of diverse application scenarios, proactive capital investments, and strategic corporate positioning all underlined the AI industry's immense potential and fueled a robust growth trajectory.

LLM Sector

Among AI's specialized technical subfields, the LLM sector was increasingly prominent within the broader AI industry and was ultimately one of the most strategically significant areas of artificial intelligence. Focused on researching, developing, applying, and commercializing large-scale pre-trained LLMs, these typically relied on Transformer architecture and leveraged large-scale unsupervised or semi-supervised learning approaches to acquire linguistic knowledge from massive amounts of textual data, sourced from the internet, which enabled them to attain accurate capabilities in natural language understanding and

38 "Artificial Intelligence," Wikipedia, last modified March 12, 2025, https://en.wikipedia.org/wiki/Artificial_intelligence, accessed March 2025.

39 "Artificial Intelligence Market Size | Industry Report, 2030," Grand View Research, March 2025, https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

generation. Representative models included the GPT series, LLaMA, Claude, Gemini, DeepSeek, and other cutting-edge tech innovations.

Since language formed the fundamental medium of human communication and knowledge transfer, the emergence of LLMs enabled AI to comprehend human language, interact seamlessly and collaborate with humans, significantly expanding the application boundaries and commercial value of artificial intelligence. LLM technology not only improved AI's performance in content generation, information retrieval, intelligent customer service, decision-making support and creative assistance, but was a crucial milestone toward achieving general artificial intelligence. It was not only a flagship achievement but a competitive battleground for the leading global AI companies, representing their research capabilities, technological ecosystem strategies, and commercialization strengths. To a significant extent, the development level of LLMs symbolized their competitiveness and influence in the broader AI landscape.

The upstream segment of the LLM market primarily provided the foundational technologies – the hardware, chips, cloud computing platforms, and data. The LLM industry itself was situated midstream, its main products comprising closed-source and open-source platforms. Downstream, LLMS were applied across various fields such as finance, healthcare, education, industry, gaming, and law. Applications mainly took the form of search engines, conversational AI, solution generation, assisted editing, enterprise services, and data analysis. Its primary role was to be integrated into products such as chatbots, intelligent customer service systems, automated writing tools, language translation services, and intelligent recommendation systems.

According to Grand View Research, the market for global generative AI was valued at USD 16.87 billion in 2024 and was projected to see a CAGR of 37.6% from 2025 to 2030. The more narrowly defined global LLM market was estimated at USD 5,617.4 million in 2024 and was projected to achieve a CAGR of 36.9% from 2025 to 2030 [see **Exhibit 3**]. Growth would be driven by the integration of zero human intervention features in training systems (self-supervised learning), which significantly enhanced efficiency. These allowed models to learn and adapt autonomously without the need for continuous manual oversight, thereby reducing the time and resources required. They facilitated scalability by enabling LLMs to handle increasing data volumes and workloads with ease.[40]

In June 2023, Databricks, Inc., a U.S.-based software company, completed the USD 1.3 billion acquisition of MosaicML, a provider specializing in LLMs and model-training software. By integrating MosaicML's models, training and inference technologies into its Lakehouse platform, Databricks would enable companies to build and deploy generative AI applications more effectively. Henceforth, more mergers and acquisitions were expected to take place, open-source and inclusive AI would gain increasing attention, and self-supervised learning, which made model training more efficient, would be widely adopted.

40 "Large Language Models Market Size, Share & Trends Analysis Report By Application (Customer Service, Content Generation), By Deployment (Cloud, On-premise), By Industry Vertical, By Region, And Segment Forecasts, 2025 - 2030," Grand View Research, https://www.grandviewresearch.com/industry-analysis/large-language-model-llm-market-report, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

By Region

North America dominated the large language model (LLM) market, with the largest revenue share of 32.1% in 2024. The United States accounting for the region's largest market revenue share[41] thanks to its advanced technological infrastructure and the presence of major companies focused on AI. Continuous investment in AI innovation and research drove widespread adoption across industries such as finance, healthcare, and retail. Government and private organizations alike strongly supported AI R&D, further accelerating the growth of LLMs. The US was the hub of LLM innovation, with Google LLC, Microsoft, and OpenAI pushing the boundaries of AI research. A range of industries adopted LLMs to improve efficiency, automate processes, and enhance customer experiences. The country's regulatory framework and investment in AI infrastructure created a favorable environment for their development.[42]

The European LLM market was emerging as a strong force, with the UK, Germany and France actively working to integrate AI into various sectors. The European Union's focus on data privacy and AI ethics shaped the market's development and adoption. European companies leveraged LLMs to streamline operations and improve customer experiences in industries such as manufacturing, retail, and healthcare. The region's growing emphasis on digital transformation was expected to drive further development of LLMs in years to come.[43]

The Asia-Pacific market experienced rapid growth, with China, Japan and India investing heavily in AI technologies. China, in particular, aimed to become a global AI leader, dedicating substantial resources to AI research and LLM integration. Japan utilized LLMs to enhance automation in robotics and manufacturing. India's expanding tech industry and the increasing use of LLMs in customer service and education contributed to the region's strong growth trajectory.[44]

Main Players in the AI Industry (and their LLMs)[45]

The following comparison focuses mainly on LLMs and chatbots based on various LLMs, representing the sector in which DeepSeek operated. As of 2025, vendors competed primarily on the performance and cost-effectiveness of their LLMs and chatbots, rather than other aspects. The comparative data used in the analysis comes from the widely recognized platform, LMArena.[46] [see **Exhibit-4** and **Exhibit-5**]

DeepSeek-R1

DeepSeek-R1 was released in January 2025. By March 2025 it was ranked sixth among 211 models on the LMArena leaderboard. Based on its performance across specific tasks, it tied third in Hard Prompts and in Coding, tied first in Math, tied third in Creative Writing, tied third in Instruction Following, tied sixth in Longer Query, and tied second in Multi-Turn tasks. In terms of cost, DeepSeek-R1 was one of the most

41 "Artificial Intelligence Market Size | Industry Report, 2030," Grand View Research, March 2025, https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market, accessed March 2025.
42 "Artificial Intelligence Market Size | Industry Report, 2030," Grand View Research, March 2025, https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market, accessed March 2025.
43 Ibid.
44 Ibid.
45 In presenting this competitive landscape, the authors relied on the data available at the time of writing. However, given that the competition in the LLM and chatbot sector was extremely intense, future conditions could potentially change dramatically.
46 "LM Arena Leaderboard," LM Arena, March 2025, https://lmarena.ai/?leaderboard=, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

cost-efficient LLMs in the world at the time, with an input cost of $0.55 per million tokens[47] and an output cost of $2.19 per million tokens.[48] [49]

OpenAI

OpenAI, Inc., an American AI research organization founded in December 2015, had headquarters in San Francisco, California. Its stated goal was to develop "safe and beneficial" artificial general intelligence (AGI), which it defined as highly autonomous systems capable of outperforming humans across most economically valuable tasks. OpenAI emerged as a prominent leader during the AI boom, known for developing the GPT series of LLMs, the DALL-E text-to-image generation models, and a text-to-video model called Sora. The release of ChatGPT catalyzed public interest in generative AI technologies.

ChatGPT, an advanced generative AI chatbot created by OpenAI, was released in 2022. It was powered by the GPT-4o LLM and designed to produce conversational responses resembling human communication. Users could tailor interactions by adjusting factors such as length, format, tone, detail and language. ChatGPT was seen as driving significant investments and public attention to artificial intelligence. Critics raised concerns that tools like ChatGPT might replace human intelligence, encourage plagiarism, or spread misinformation.

By January 2023 ChatGPT was the fastest-growing consumer software application, surpassing 100 million users within two months. Its introduction triggered the release of competing AI tools, including Gemini, Claude, Llama, Ernie, and Grok. Microsoft launched Copilot, which was initially based on GPT-4. In May 2024, Apple announced a collaboration with OpenAI to integrate ChatGPT into its operating systems through the Apple Intelligence feature. By July 2024, the ChatGPT website was ranked as one of the top 10 most-visited websites worldwide.

GPT-4.5-Preview was released in February 2025. By March it tied for first place among 211 models on the LMArena leaderboard. Based on its performance across specific tasks, it tied first in Hard Prompts, Coding, Math, Creative Writing, Instruction Following, Longer Query, and Multi-Turn tasks. In terms of cost, GPT-4.5-Preview was the most expensive LLM to train in the world at the time, with an input cost of $75 per million tokens and an output cost of $150 per million tokens – 280 times the input cost and 150 times the output cost of DeepSeek V3.[50]

Google

With unparalleled access to massive amounts of diverse, real-time data collected daily from its various services (Search, YouTube, Gmail, Google Maps, and Android) Google AI was able to train reliable models like the Gemini series, significantly enhancing their predictive and generative capabilities. Google had pioneered research in AI, developing groundbreaking innovations like the Transformer architecture, the TensorFlow framework, and achievements in reinforcement learning. Its powerful computational

47 Tokens are the basic text units that language models process, typically representing parts of words, whole words, or characters, which are counted for both input and API pricing calculations.

48 Ibid.

49 "How Should We View DeepSeek's $5.576 Million Training Cost?," Sina Technology, February 8, 2025, https://finance.sina.com.cn/tech/roll/2025-02-08/doc-ineitvcc9059877.shtml, accessed March 2025.

50 "LM Arena Leaderboard," LM Arena, March 2025, https://lmarena.ai/?leaderboard=, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

infrastructure included proprietary Tensor Processing Units (TPUs), extensive data centres, and cloud computing resources (Google Cloud Platform), which allowed efficient training and deployment of large-scale AI models. The integrated hardware-software stack reduced latency, improved efficiency, and lowered operational costs.

What could not be ignored was that Google invested significant financial resources in AI R&D, strategic acquisitions and top-tier talent recruitment. It attracted and retained some of the world's leading AI researchers and engineers by offering competitive salaries, ample resources, and an appealing environment for innovation and career growth.

Gemini, a series of multimodal large language models developed by Google DeepMind, was designed as the successor to LaMDA as part of the next generation of the PaLM 2 series. It included Gemini Ultra, Gemini Pro, and Gemini Nano, officially launched on December 6, 2023 to compete with OpenAI's GPT-4.

On December 11, Google formally released the latest version, Gemini 2.0, which had advanced multimodal capabilities that supported native image and multilingual audio output, as well as native tool usage. This allowed the model to understand and process various input types, including text, images, video and audio, while generating corresponding outputs. In addition, Gemini 2.0 integrated powerful tools such as Google Search, code execution, and third-party user-defined functions, which significantly expanded its range of applications.

Until March 2025, Gemini-2.0-Flash-Thinking-Exp-01-21 was its most advanced and powerful model, and ranked third among the 211 models listed on the LMArena leaderboard. Across different tasks, it ranked second in the "Hard Prompts" category, tied third in "Coding," tied first in "Math," tied second in "Creative Writing", "Instruction Following," "Longer Query" and in "Multi-Turn" interactions. The open-source Gemini series had significant cost advantages.

xAI

On July 14, 2023, Elon Musk formally announced the creation of xAI, a new artificial intelligence company, via his social media account. A Twitter Spaces event was held the same day to discuss the company's vision and objectives. Unlike conventional AI companies, xAI was positioned to work in collaboration with Musk's other ventures, including Tesla and Twitter (later rebranded X). One of xAI's primary goals was to develop AI models capable of advanced logical reasoning and problem-solving, aiming to surpass existing generative AI systems in capability and reliability. Musk described xAI's mission as an effort to build a "maximally curious" AI system, one that would seek to understand the universe in depth while maintaining alignment with human values.

On January 3, 2025, Musk announced that xAI, had completed the pre-training of its latest generative AI chatbot, Grok 3, with computing power ten times greater than its predecessor Grok 2. Grok 3 adopted the Mixture of Experts (MoE) architecture, consisting of eight expert models, with each data unit (token) being processed by two experts. Each processing task involved up to 86 billion activation parameters. Additionally, Grok 3 was based on the Transformer architecture, featuring 64 layers and utilizing 48 attention mechanism units for query processing and 8 key/value attention mechanism units.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

By March 2025, it was ranked tied for first place among 211 models on the LMArena leaderboard, tied third in Hard Prompts and in Coding, tied first in Math and in Creative Writing, tied first in Instruction Following and in Longer Query, and tied second in Multi-Turn tasks. While the Grok models did not disclose specific input and output costs, during the launch of Grok 3, Musk said that training the model required 200,000 Nvidia GPUs, which were undoubtedly a significant expense.[51]

Alibaba

Alibaba's occupied an industry-leading position in this AI wave thanks to its forward-thinking approach. Around 2015, its businesses Taobao and Alipay experienced rapid growth, as did the volume of data the company needed to handle daily.

In March 2021, Alibaba collaborated with Tsinghua University to launch the industry's largest Chinese multimodal pre-trained AI model, M6. The model had over 100 billion parameters and was capable of understanding and generating both text and images, surpassing human efficiency in image design tasks. It applied to areas such as product design, information retrieval, robotic dialogue, and creative writing. OpenAI was not, at that time, so widely known, so Alibaba's breakthroughs in multimodal pretraining demonstrated it was not merely an imitator but an innovator. When ChatGPT emerged at the end of 2022, Alibaba was able to swiftly launch its own large-scale model, Tongyi Qianwen (Qwen). From September 2023, Alibaba was explicitly "AI-driven" as the strategic focus of the entire group.

Implementation followed a two-part approach: "Public cloud first" and "Open-source and openness". This was underpinned by its innovation philosophy – first, solve problems faced by industries and society, then achieve growth through industry and societal progress.

Large models continued to evolve technologically. Yet for industries and society, the primary issue was not the technology but the fact that these models were too expensive to use, and even more expensive to develop. For AI to be embraced by society, it was crucial to reduce computing costs and lower the barriers to its development and use; otherwise, large-scale models would remain a luxury reserved for a small number of tech giants. Like DeepSeek's choice to go open-source, Alibaba believed the solution to reducing computing costs was to focus on developing the public cloud, and the solution to lowering the barriers was open-source and openness.

In terms of open-source and openness, Alibaba pushed for comprehensive open-sourcing across all model sizes and modalities. As of that time, the Tongyi Qianwen series of open-source models had surpassed 40 million global downloads. Worldwide developers had created over 80,000 derivative models based on Tongyi Qianwen, a number comparable to Meta's Llama series, putting Alibaba among the global leaders. Leveraging Alibaba's open-source models, developers and enterprises no longer needed to train large models themselves. Instead, they could customize existing models, drastically reducing the major expenses associated with AI application and deployment, which was critical for widespread adoption.

After more than a year of implementing its AI-driven strategy, Alibaba achieved the expected outcomes along both strategic paths – "Public cloud first" and "Open-source and openness." These also significantly boosted Alibaba Cloud's performance. On November 15, 2024, its latest quarterly earnings report stated

51 "LM Arena Leaderboard," LM Arena, March 2025, https://lmarena.ai/?leaderboard=, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

that Alibaba Cloud's revenue growth accelerated for the fourth consecutive quarter. Quarterly adjusted EBITA profit reached 2.661 billion yuan, representing a year-over-year increase of 89%.

On January 29, 2025, Alibaba released a new version of its Qwen2.5 AI model, reportedly superior to DeepSeek-V3. An official statement was published on the Alibaba Cloud division's WeChat account: "The performance of Qwen2.5-Max has almost entirely surpassed GPT-4o, DeepSeek-V3, and Llama-3.1-405B," referring to OpenAI and Meta's most advanced open-source AI models. Qwen2.5-Max ranked 9th among the 211 models listed on the LMArena leaderboard in March 2025. It ranked tied 5th in "Hard Prompts," tied 7th in "Coding," 5th in "Math," 8th in "Creative Writing," tied 8th in "Instruction Following," tied 7th in "Longer Query," and tied 7th in "Multi-turn" interactions. Qwen2.5-Max had moderate cost-effectiveness.[52]

The Future of DeepSeek: A Game Changer?

DeepSeek defied several conventional beliefs. The first was "The US leads everything." For a long time, America had been the leader in AI technology innovation, while Chinese companies had been followers, excelling at expanding application scenarios rather than achieving breakthroughs. However, a Chinese team had now managed to develop a "China-made" AI model that was both high-quality and cost-effective, akin to consumer electronics, shattering the stereotype that technological innovation could only emerge from the US. Turing Award winner Yann LeCun took the opportunity to criticize what he called "Silicon Valley arrogance", rooted in the belief that people there had a monopoly on good ideas and dismissed innovations outside the group as the result of "cheating."

The second was "Massive capital investment and market monopolization lead to excessive profits." The launch of the DeepSeek LLM sent shockwaves through the US stock market, with Nvidia's stock plummeting nearly 17% in a single day, wiping out $595 billion in market value—equivalent to 4.3 trillion RMB—the largest single-day drop in US stock market history. This was not only driven by fears that chip demand might decrease due to DeepSeek's lower training costs, but also by the impact on the traditional logic of American tech capital that massive investment in an industry would provide a few oligopolistic companies with a dominant financial advantage, enabling them to achieve breakthroughs and establish a monopolistic market.

In the AI industry, where computing costs were extraordinarily high, this logic was amplified. For instance, the "Stargate" project, jointly promoted by OpenAI, SoftBank's Masayoshi Son and Middle Eastern funds, aimed to invest $500 billion over four years to create a "global computing power center" and secure an advantage through large-scale investment. The emergence of DeepSeek LLM delivered a heavy blow to this capital-driven logic.

In the AI field, massive investment and monopolization would no longer guarantee lasting dominance. Leading the market might not require such enormous funds, and even with such investments, closed-source models might still fail to secure monopolistic control. If this fundamental tenet started to be questioned, capital would need to rethink its investment strategy.

52 "LM Arena Leaderboard," LM Arena, March 2025, https://lmarena.ai/?leaderboard=, accessed March 2025.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

The third was “Closed-source models always lead the way.” On the journey toward AGI, the rise of DeepSeek not only dismantled the old closed-source monopoly but redefined the rules of competition through its open-source and democratized approach. From "US-led" to "China-driven innovation," from "capital-intensive" to "accessible to all," and from "closed-source monopoly to open collaboration", these fundamental shifts were shaping a new order for AI development.

At the same time, DeepSeek would face the dual challenge of advancements in model technology and geopolitical obstacles. Although DeepSeek gave a strong boost to China's AI industry, it did not resolve the current issues in the foundational technology of large language models. Problems like “hallucination” remained difficult to address and might require paradigm-shifting innovations to overcome. Multimodal models faced even greater challenges, such as poor instruction-following performance for images and lack of detail control. While video generation models had progressed rapidly over the past six months, issues like instruction-following and consistency still left room for improvement, and the costs remained high. Models like DeepSeek V3, which significantly reduced costs, were urgently needed.

While the emergence of DeepSeek gave Chinese companies greater confidence to pursue innovation and served as a wake-up call for the US, the technological “cold war” between the two was expected to intensify. For example, restrictions on high-performance chips might be tightened even further. Could DeepSeek continue its strategy and ambition? Would its passion for technology take it beyond being a disruptor to permanently changing the rules of the game in the AI industry and the LLM sector? How would China’s dark horse achieve profitability in the future?

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

INSEAD The Business School for the World®

Exhibit 1

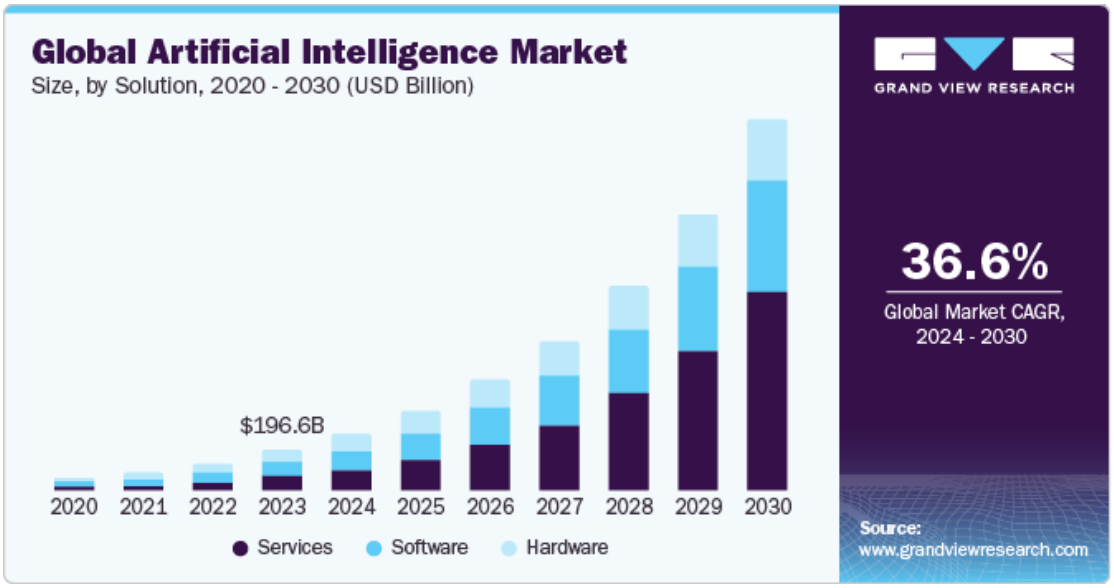
Timeline of the Launch of DeepSeek LLMs



Source: Organized by the author by public information.

Exhibit 2

Global AI Market Value



Source: www.grandviewsearch.com https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market.

Copyright © INSEAD

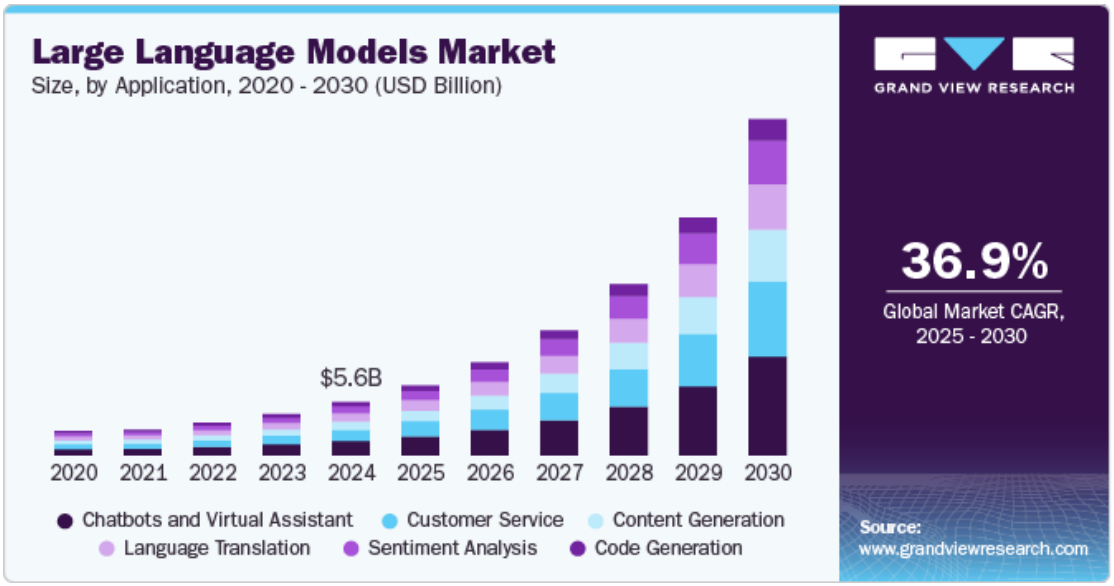
This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

INSEAD The Business School for the World®

Exhibit 3

Global LLM Market Value



Source: www.grandviewsearch.com https://www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-market

Exhibit 4

LLM Rankings on LMArena (March 2025)

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License
1	2	Grok-3-Preview-02-24	1412	+8/-10	3364	xAI	Proprietary
1	1	GPT-4.5-Preview	1411	+11/-11	3242	OpenAI	Proprietary
3	5	Gemini-2.0-Flash-Thinking-Exp-01-21	1384	+6/-5	17487	Google	Proprietary
3	3	Gemini-2.0-Pro-Exp-02-05	1380	+5/-6	15466	Google	Proprietary
3	2	ChatGPT-4o-latest (2025-01-29)	1377	+5/-4	17221	OpenAI	Proprietary
6	3	DeepSeek-R1	1363	+8/-6	8580	DeepSeek	MIT
6	10	Gemini-2.0-Flash-001	1357	+6/-5	13257	Google	Proprietary
7	3	o1-2024-12-17	1352	+4/-6	19785	OpenAI	Proprietary
9	10	Qwen2.5-Max	1336	+7/-5	11930	Alibaba	Proprietary
9	7	o1-preview	1335	+4/-3	33167	OpenAI	Proprietary
9	10	o3-mini-high	1329	+8/-6	9102	OpenAI	Proprietary
11	13	DeepSeek-V3	1318	+5/-4	22007	DeepSeek	DeepSeek

Source: https://lmarena.ai/

Copyright © INSEAD

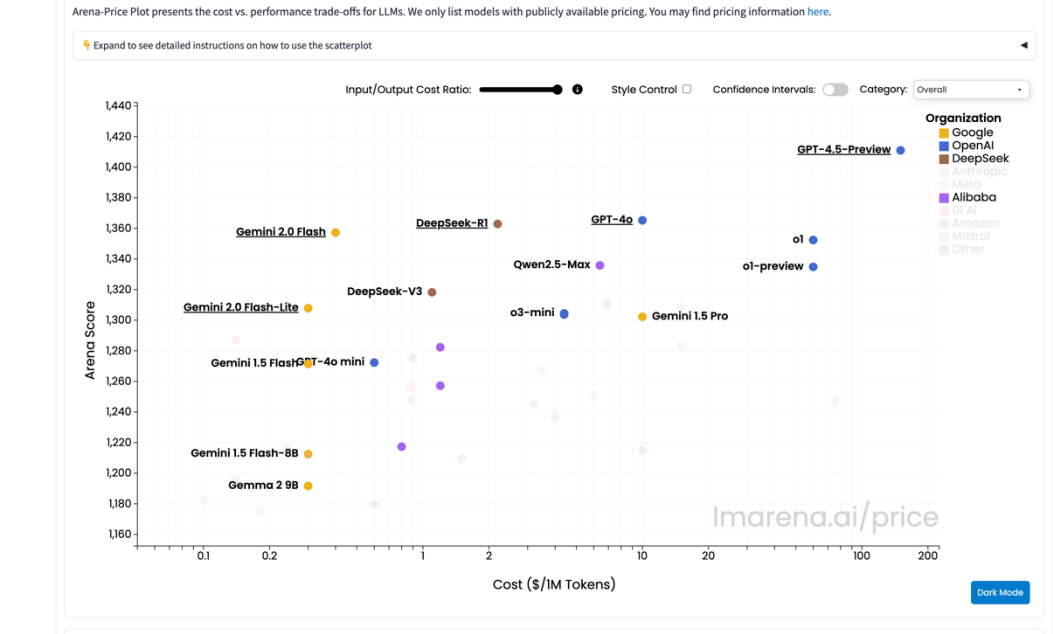
This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

INSEAD The Business School for the World®

Exhibit 5

Cost Overview of Google's OpenAI's, Alibaba's and DeepSeek's LLMs



Source: https://lmarena.ai/

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.